Multiplexing

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What is Multiplexing?

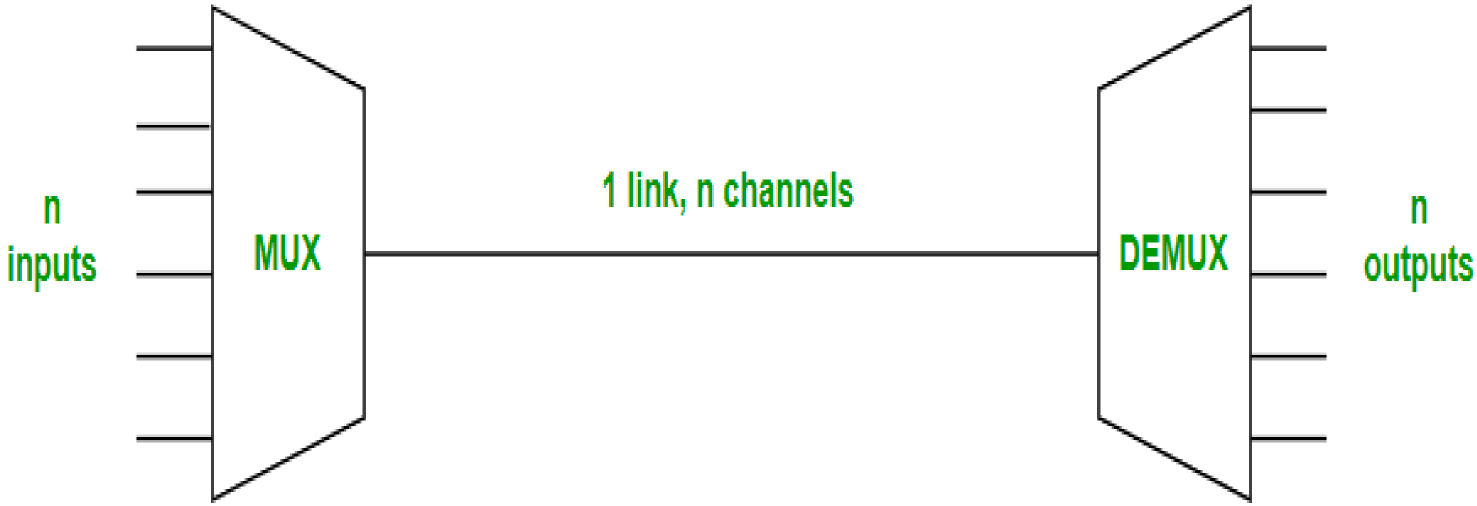
- Multiplexing is a technique that **combines multiple signals into a single signal** for transmission over a shared communication medium.

- It **improves bandwidth utilization** and allows simultaneous data transmission.

Why Use Multiplexing?

- Efficient utilization of bandwidth
- Supports simultaneous multiple calls/data streams
- Used in satellite, telecom, and computer networks
- Increases capacity and throughput

Basic Concept



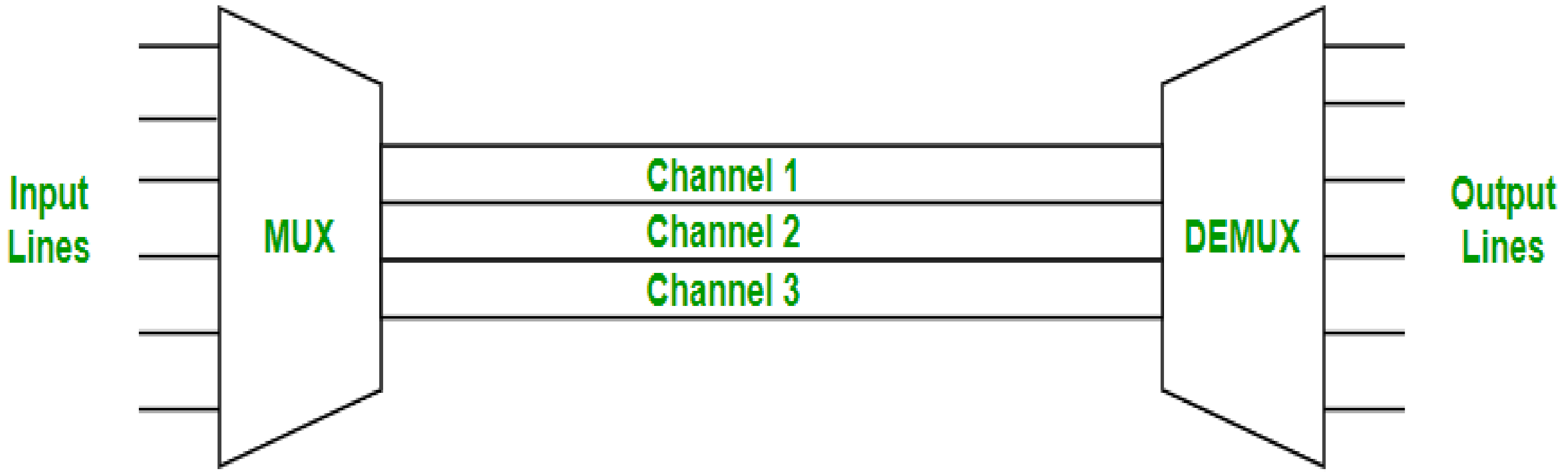
- **MUX combines signals; DEMUX separates them at receiver.**

Types of Multiplexing

1. Frequency Division Multiplexing (FDM)
2. Time Division Multiplexing (TDM)
   - Synchronous TDM
   - Statistical TDM
3. Wavelength Division Multiplexing (WDM)
   - CWDM
   - DWDM
4. Code Division Multiplexing (CDM)

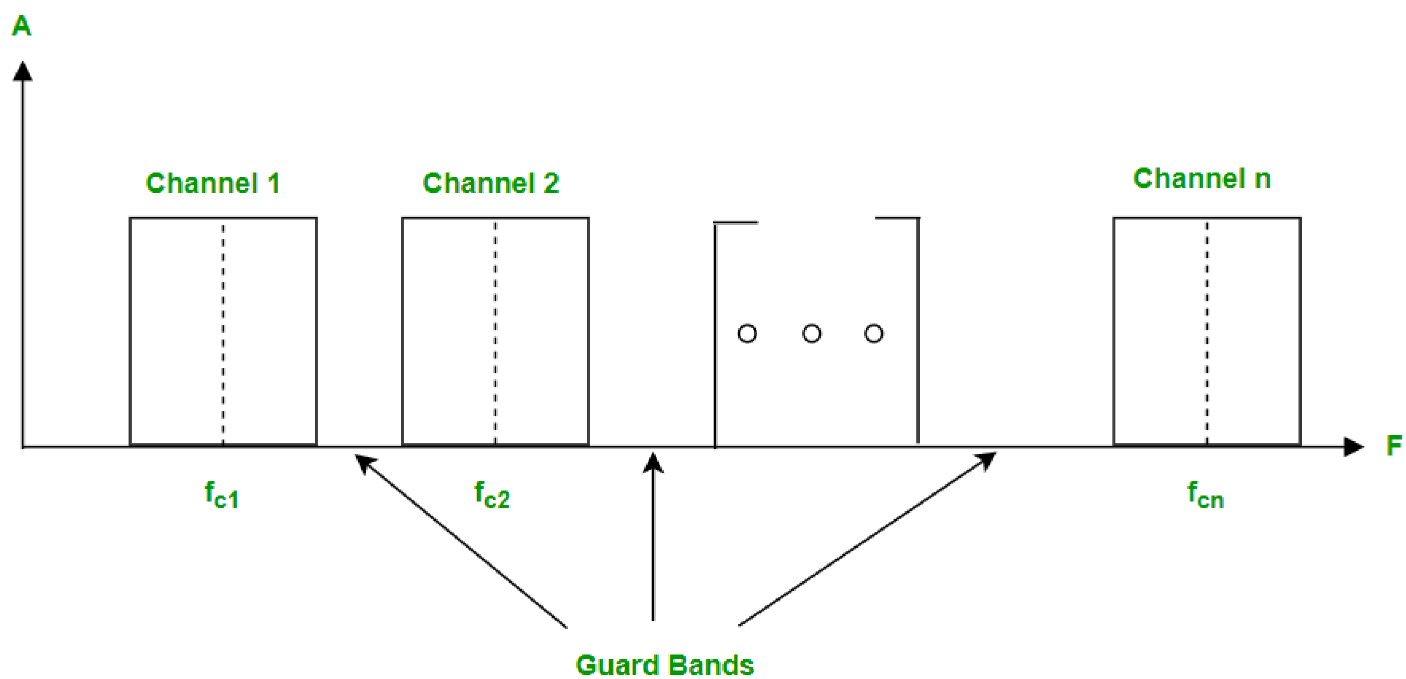
Frequency Division Multiplexing (FDM)

- FDM divides **total bandwidth into smaller frequency bands.**
- Each signal uses its own frequency simultaneously.
- Used in **radio/TV**
- Requires guard bands to prevent interference


Input
Lines
MUX
Channel 1
Channel 2
Channel 3
DEMUX
Output
Lines

FDM

- In order to prevent the inter-channel cross talk, unused strips of bandwidth must be placed between each channel. **These unused strips between each channel are known as guard bands.**

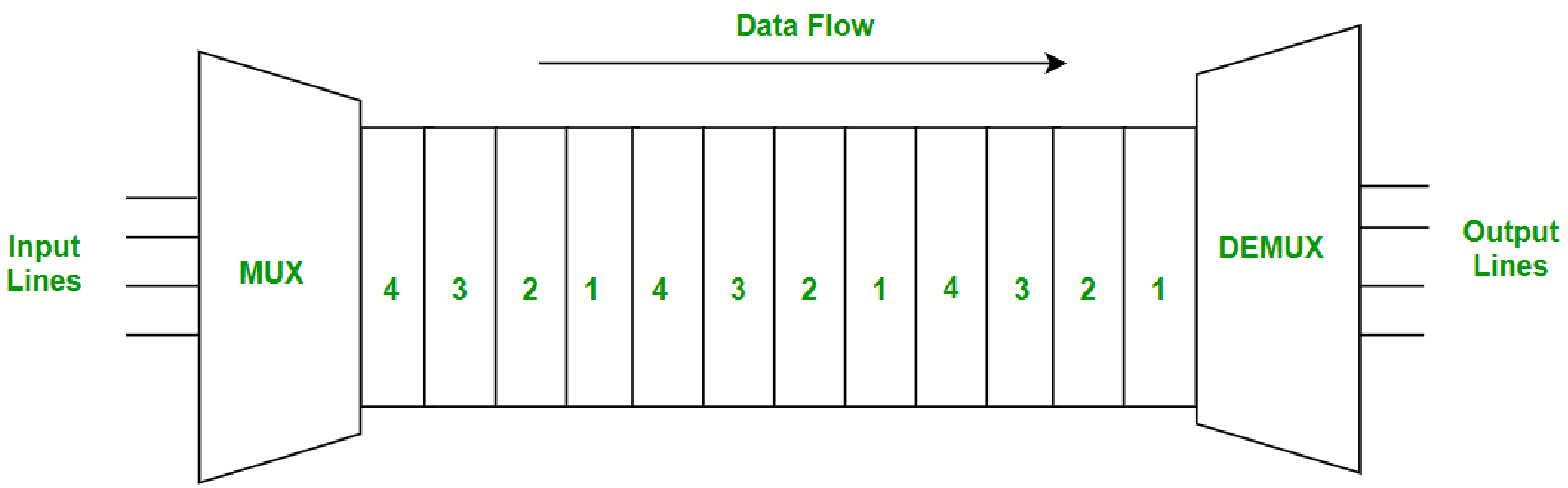


Time Division Multiplexing (TDM)

- TDM divides transmission time into slots.
- **All signals share same frequency but transmit in different time intervals**.

Types:

- Synchronous TDM
- Statistical TDM



Time Division Multiplexing

Synchronous TDM

- Each device gets a fixed time slot in every frame.
- Slots are transmitted even if the device has no data.

Advantage: Simple

Drawback: Wastage of bandwidth

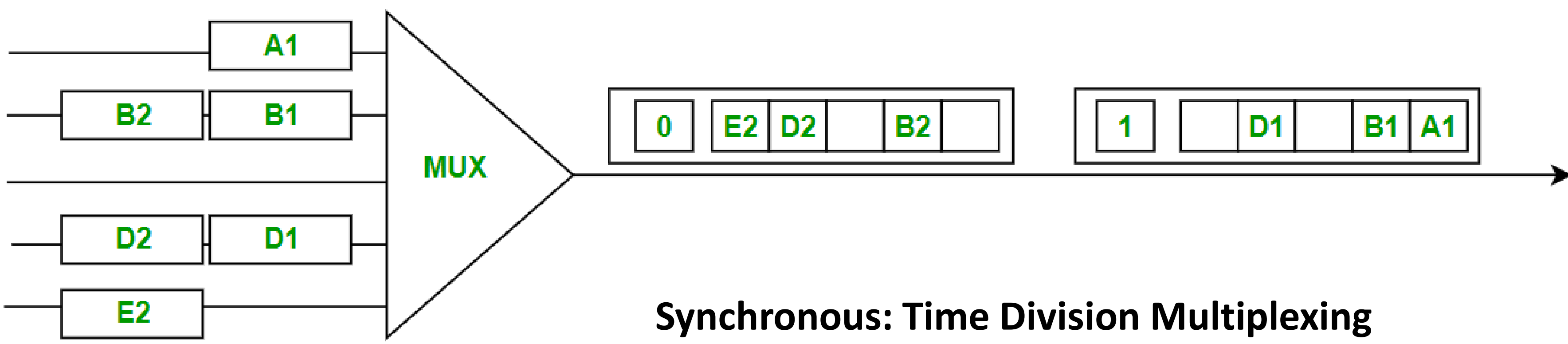
- Example: Traditional telephone systems

Synchronous: Time Division Multiplexing

- In synchronous TDM, **every device which is present in this has given the same time slot to transmit data.**
- It does not consider whether the device contains data or not.
- **The devices place their data on the link when their time slots arrive, if somehow any device does not contain data its time-slot remains empty.**
- There are various kinds of time slots that are organized into frames and each frame consist of one or more time slots dedicated to each sending device.

SYNCHRONOUS TDM

- **Time slots are grouped into frames.**
- One frame consists of one cycle of time slots.
- Synchronous TDM is not efficient because if the input frame has no data to send, a slot remains empty in the output frame.
- In this, we need to mention the synchronous bit at the beginning of each frame for frame boundary detection, Slot alignment and Timing



Synchronous: Time Division Multiplexing

Statistical TDM

- Time slots are allocated dynamically only when devices have data.
- Statistical TDM is a type of Time Division Multiplexing where the **output frame collects data from the input frame till it is full not leaving an empty slot like in Synchronous TDM**
- In this, we need to include the address of each particular data in the slot that is being sent to the output frame.
- **Statistical TDM is a more efficient type of time-division multiplexing as the channel capacity is fully utilized** and improves the bandwidth efficiency.

STATISTICAL TDM

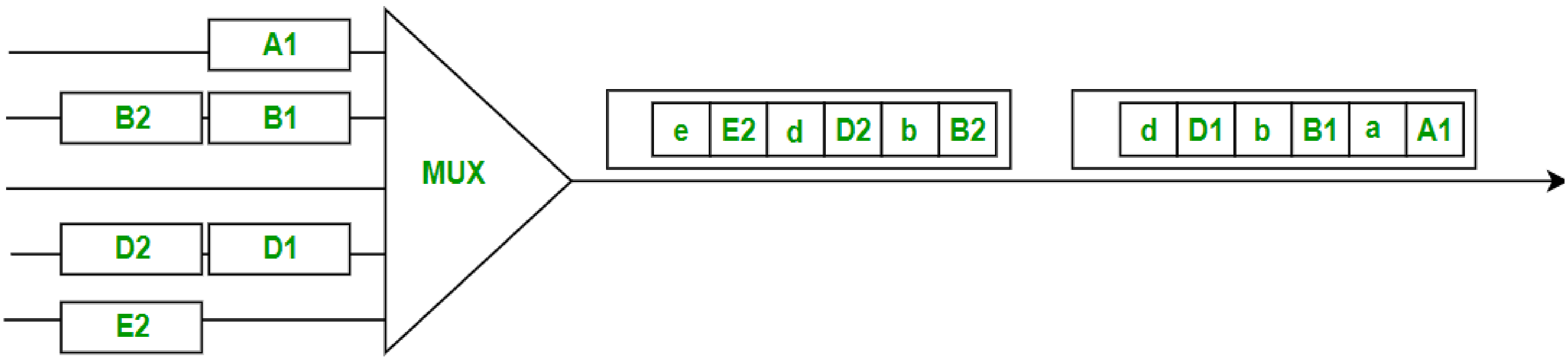
Advantage

- Better bandwidth utilization

Drawback

- No empty slot wastage
- Requires addressing info

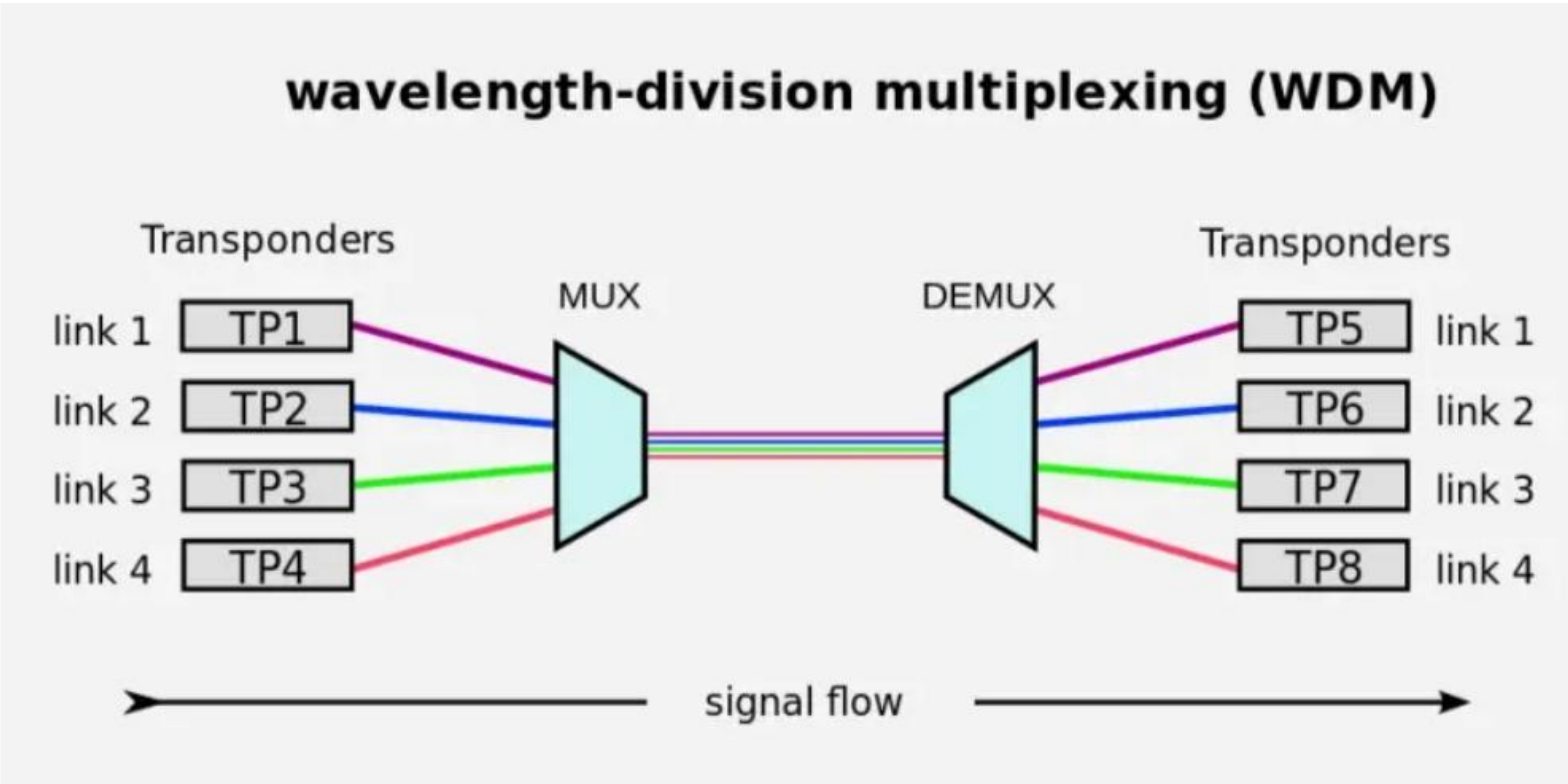
Example: Packet-switched networks



Statistical: Time Division Multiplexing

Wavelength Division Multiplexing (WDM)

- Optical multiplexing technique in fiber optics.
- Multiple signals are transmitted using different wavelengths of light.
- Wavelength Division Multiplexing (WDM) is a multiplexing technology used to **increase the capacity of optical fiber by transmitting multiple optical signals simultaneously over a single optical fiber, each with a different wavelength.**


wavelength-division multiplexing (WDM)
Transponders
Transponders
MUX
DEMUX
link 1
TP1
link 2
TP2
link 3
TP3
link 4
TP4
TP5
link 1
TP6
link 2
TP7
link 3
TP8
link 4
signal flow

WDM

WDM

- Each signal is carried on a different wavelength of light, and the **resulting signals are combined onto a single optical fiber for transmission.**
- At the receiving end, **the signals are separated by their wavelengths, demultiplexed and routed to their respective destinations**.
- It is used in telecommunications, cable TV, ISPs, and data centers for high-speed, long-distance data transmission.

Types of WDM

- **Dense Wavelength Division Multiplexing (DWDM)** is used to multiplex a large number of optical signals onto a single fiber, typically up to 80 channels with a spacing of 0.8 nm or less between the channels.

- **Coarse Wavelength Division Multiplexing (CWDM)** is used for lower-capacity applications, typically up to 18 channels with a spacing of 20 nm between the channels.

Code Division Multiplexing (CDM)

- Multiple signals share same frequency band using unique codes.
- Used in CDMA mobile systems.
- Each user has full bandwidth with unique spreading code.

Advantages of Multiplexing

- Efficient bandwidth utilization
- Increased transmission capacity
- Cost-effective
- Supports multiple users simultaneously

Disadvantages of Multiplexing

- Complex design
- Synchronization challenges
- Signal degradation/interference
- Added overhead in statistical methods